Fast and reliable smoothness-indexed penalized least squares across difference orders: numerical algorithms, grid reuse and computational scaling

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ABSTRACT

Finite-difference penalized least-squares (PLS) trends can be parameterized by a normalized, interpretable smoothness index $s \in [0,1]$ rather than the scale-dependent penalty λ. Efficient use of this coordinate requires more than a fast linear solver: exact smoothness inversion, conditional effective degrees of freedom, leave-one-out leverage values, criterion evaluation and repeated-grid reuse can dominate complete computational time. We present an algorithmic analysis for difference orders $d = 1, \ldots, 6$ and fixed zero or nonzero means of penalized differences, including an exact banded selected-inverse formulation for full conditional CV, GCV, AICc, residual BIC and Gaussian marginal-likelihood evaluation. We define four crossed workloads: one or many smoothness candidates evaluated for one or many independent series. Nine full-criterion pipelines—naive dense inversion, spectral SVD, banded Cholesky and six explicitly costed banded/sparse/iterative hybrids—form a staged, resource-gated evaluation design over seven sample sizes through 131072, four positive smoothness grids through 100000 candidates, two observation magnitudes and fixed reference means. This large-scale factorial is a *planned experiment, not an empirical result.* The existing two-seed, 80640-row Windows timing corpus supplies a historical baseline: sharing one exactly matched smoothness-grid translation across five criteria reduced measured elapsed time by factors of 2.67 at $d = 3$ and 2.82 at $d = 2$ for specified $n = 48$ workflows. We distinguish these host-specific observations from theoretical complexity and future results, require numerical validity before timing comparisons and preserve $s = 1$ as an exact constrained endpoint.

1. Introduction and smoothness coordinate

Quadratic finite-difference penalized least squares (PLS) has a long history in Whittaker–Henderson smoothing (Whittaker, 1923; Guerrero, 2007). The penalty λ is mathematically convenient, but the same numerical value need not express the same smoothness for different difference orders or sample lengths. Guerrero's trace-based smoothing measure (Guerrero, 2008) and its use with CV, GCV, AICc and BIC (Cortés-Toto, Guerrero and Reyes, 2017) are established precedents. We do not claim to invent the smoother or its tuning criteria; instead, we investigate the *computational cost of an interpretable bounded coordinate and the reuse of its mapping across many evaluations.*

For $y \in \mathbb{R}^n$, difference order $1 \le d < n$, and fixed mean of the penalized differences μ, let $D_d \in \mathbb{R}^{(n-d)\times n}$ denote forward differences and define

$$\widehat{\tau}_{\lambda,\mu} = \arg\min_{\tau}\{\|y-\tau\|_2^2 + \lambda\|D_d\tau - \mu\mathbf{1}\|_2^2\}. \quad (1)$$

Put $r = n - d$, $Q_d = D_d^\top D_d$, and $H_\lambda = (I + \lambda Q_d)^{-1}$. A polynomial particular solution is $p_{\mu,t} = \mu\binom{t-1}{d}$, and $\widehat{\tau} = p_\mu + H_\lambda(y - p_\mu)$. The penalty has a d-dimensional polynomial nullspace, so $\mathrm{tr}(H_\lambda) \to d$ as $\lambda \to \infty$.

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Write $\delta_1, \ldots, \delta_r > 0$ for the positive eigenvalues of Q_d. We use the normalized smoothness index

$$S_d(\lambda) = \frac{n - \mathrm{tr}(H_\lambda)}{n-d} = \frac{1}{r}\sum_{j=1}^{r}\frac{\lambda\delta_j}{1+\lambda\delta_j}. \quad (2)$$

Each $\delta_j > 0$ implies $S_d'(\lambda) = r^{-1}\sum_j \delta_j/(1+\lambda\delta_j)^2 > 0$, hence S_d is continuous, strictly increasing, and maps $[0,\infty]$ onto $[0,1]$. Every interior s has a unique penalty $\Lambda_{n,d}(s)$. The map depends only on (n,d), *not* on observations or the fixed reference mean μ. Therefore it can be cached across different series, criteria and μ values. The denominator $n-d$ normalizes an established trace-based index; it is not a new statistical model.

1.1. Shift-free full spectral reconstruction

Let $D_d = U\,\mathrm{diag}(\sigma)V^\top$ be a reduced SVD over the r positive singular modes and write $w_j = \lambda\sigma_j^2/(1+\lambda\sigma_j^2)$. A numerically useful identity is

$$\widehat{\tau}_{\lambda,\mu} = y - V\big[w \odot (V^\top y) - \mu(w \oslash \sigma) \odot (U^\top\mathbf{1})\big], \quad (3)$$

where multiplication and division are componentwise. It avoids forming the potentially enormous polynomial p_μ at high d and costs $O(nr)$ for a complete curve once the SVD is available. A separate high-precision

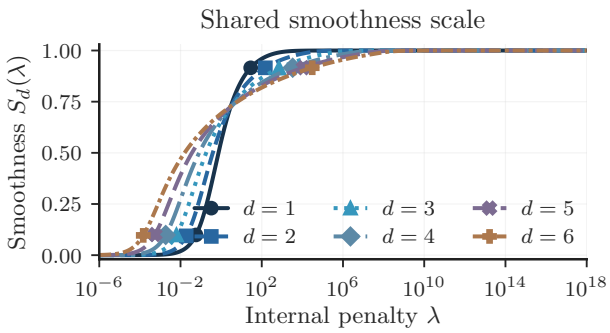


Figure 1: Smoothness versus penalty for $n = 48$, orders $d = 1, \ldots, 6$. The sixth-order line illustrates the mathematical mapping, not a completed Windows timing benchmark for $d = 6$.

companion manuscript evaluates its numerical accuracy. Here we study *runtime costs* without treating a numerically invalid fast method as a winner.

The exact endpoints are interpolation at $s = 0$ and the polynomial-constrained limit at $s = 1$, which is evaluated analytically rather than factorizing a matrix at $\lambda = \infty$. Figure 1 illustrates why a user benefits from a common unit-interval smoothness coordinate even though the corresponding penalty scales vary with d.

1.2. Higher-order differences and physical interpretation

When y_t represents a sampled position at equally spaced times $t_j = j\Delta t$, the rescaled finite differences $(D_d\tau)_j/(\Delta t)^d$ approximate the dth time derivative as $\Delta t \to 0$ for sufficiently smooth trajectories. Thus $d = 1$ penalizes departures from constant velocity (first differences), $d = 2$ from constant acceleration (second differences), and $d = 3$ from constant jerk (third differences). This interpretation requires that the measured variable is position and the sampling step is meaningful; for generic economic or environmental series these are merely algebraic difference orders.

At the maximum-smoothness limit, the penalty constraint is $D_d\tau = \mu\mathbf{1}$. For $\mu = 0$, the limiting fit belongs to the polynomial nullspace of D_d and has degree at most $d - 1$: $d = 1$ gives a constant, $d = 2$ a straight line, $d = 3$ a quadratic and $d = 4$ a cubic. For a *nonzero fixed* μ, the limiting fit has degree d (for example, $d = 3$ permits a cubic with constant third difference, corresponding to constant nonzero jerk for uniformly spaced positions). In the physical units of position, $\mu/(\Delta t)^3$ approximates jerk; μ alone is not a jerk value without units and spacing.

1.3. Research questions

A single fitted curve, one criterion score, a complete grid search, and R independent searches are different workloads. We distinguish: setup costs; full fitted-trend reconstruction; trace and leverage evaluation for CV/GCV/AICc/BIC/MLE; one-point evaluation; matched 17-point grids; locally optimized continuous searches; and repeated-series reuse. Spectral preprocessing may be expensive for one query yet efficient when amortized. The question is not whether s magically lowers the cost of optimizing λ, but whether the *same exact set* $\Lambda_{n,d}(s_j)$ can be calculated once, reused, and exposed to users as interpretable $s_j \in [0, 1]$.

2. Algorithmic time and storage complexity

Let n be series length, d difference order, $r = n - d$, K penalty candidates, R observed series, C scoring criteria and I root-solver evaluations per requested s. All big-O statements are arithmetic models, not measured wall-clock rates. The penalty matrix $Q_d = D_d^\top D_d$ has half-bandwidth d.

2.1. Factorization and one complete fitted curve

Full SVD of the $r \times n$ difference matrix requires $O(nr^2)$ time and $O(nr)$ storage, or $O(n^3)$ and $O(n^2)$ respectively when d is small relative to n. Dense symmetric eigendecomposition of Q_d also has cubic time and quadratic storage; forming Q_d can exacerbate spectral conditioning. After spectral setup a full fitted curve uses $O(nr)$ products. Dense general solves and dense Cholesky require $O(n^3)$ per new penalty. General banded solving and banded Cholesky can factor in $O(nd^2)$ and use $O(nd)$ storage, followed by $O(nd)$ triangular or back substitution. Thus structured methods can be substantially faster for a *single fit*; Cholesky and general banded LU must not be conflated.

2.2. Criterion evaluation is not the same as fitting

For a fixed λ, the conditional effective degrees of freedom are $\nu(\lambda) = d + \sum_j (1 + \lambda\delta_j)^{-1}$ and normalized smoothness is (2), both computed in $O(r)$ after spectral preparation. A projected observation $z = V^\top(y - p_\mu)$ costs $O(nr)$ once, after which $\mathrm{RSS}(\lambda) = \sum_j w_j(\lambda)^2 z_j^2$ and the conditional GCV, AICc, BIC and profiled Gaussian marginal likelihood require $O(r)$ spectral sums per candidate. Classical fixed-μ LOOCV also requires leverages $(H_\lambda)_{ii}$ and residuals: calculating both using stored right singular vectors costs $O(nr)$ per penalty. This is *conditional observation-weight deletion*, not rolling-origin forecast validation, and empirical plug-in $\widehat{\mu} = \mathrm{mean}(D_d y)$ would not have the same exact leave-one-out interpretation without refitting that mean.

Nonspectral methods may fit quickly while incurring additional exact EDF and leverage work. Simply solving n banded right-hand sides to obtain inverse

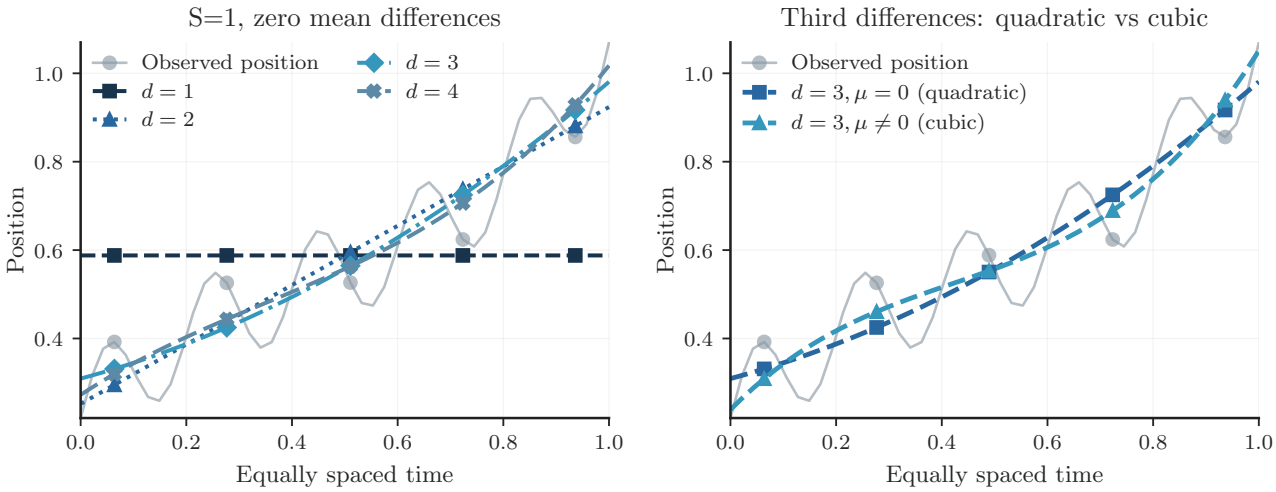


Figure 2: Deterministic illustration of limiting ($S = 1$) polynomial trends on a sample position trajectory, not a timing experiment. Left: zero-mean dth differences give polynomial degree at most $d - 1$. Right: $d = 3$, $\mu = 0$ gives a quadratic; a nonzero fixed third-difference mean gives a cubic, which corresponds to constant jerk only after scaling by the time step cubed.

Table 1
Dominant arithmetic for the implemented algorithm families. Costs are conditional on fixed μ and known finite λ. EDF and CV entries for nonspectral methods are not interchangeable with their fit cost. We state $r = n - d$ and assume a dense spectral basis is retained.

Method	Decomposition/setup	One fit	EDF/GCV	LOOCV	Memory
Fresh dense solve	$O(nd^2)$	$O(n^3)$	$O(n^3)$	$O(n^3)$	$O(n^2)$
Cached dense solve	$O(nd^2)$	$O(n^3)$	$O(n^3)$	$O(n^3)$	$O(n^2)$
Dense Cholesky	$O(nd^2)$	$O(n^3)$	$O(n^3)$	$O(n^3)$	$O(n^2)$
General banded LU	$O(nd^2)$	$O(nd^2)$	method-specific	method-specific	$O(nd)$
Banded Cholesky	$O(nd^2)$	$O(nd^2)$	method-specific	method-specific	$O(nd)$
Dense eigendecomposition	$O(n^3)$	$O(n^2)$	$O(n)$	$O(n^2)$	$O(n^2)$
Full SVD of D_d	$O(nr^2)$	$O(nr)$	$O(r)$	$O(nr)$	$O(nr)$

diagonals costs up to $O(n^2 d)$ beyond the $O(nd^2)$ factorization; specialized selected-inverse or structured algorithms can improve this (Weinert, 2007; Biessy, 2026). We distinguish the actual timed implementations from these more specialized competitors and avoid advertising a fit-only cost as an end-to-end CV-selection cost.

For dense direct methods, EDF and LOOCV may be computed using a factorization and many right-hand sides; the displayed cubic bound reflects this full operation. A properly implemented banded selected inverse is not necessarily governed by the straightforward many-right-hand-side cost. These costs exclude software-level allocation and numerical conditioning, both measured or audited separately.

2.3. How pretranslation and grid reuse change total work

With the singular values cached, one inverse $s \mapsto \lambda$ is a monotone scalar root problem in $\log \lambda$ whose function calls each take $O(r)$, so K targets with I iterations cost $O(KIr)$ beyond the SVD. Let the set $G_{n,d} = \{\Lambda_{n,d}(s_j) : 1 \leq j \leq K\}$ be cached. It depends neither on the current series nor on the fixed mean of differences. The same grid can be reused across C criteria and R new series.

For GCV/AICc/BIC/ML-style spectral selections, an illustrative amortized cost is

$$T_{\text{SVD}} = O(nr^2 + KIr + Rnr + RCKr). \tag{4}$$

The Rnr term projects each new series once; CV replaces the pertinent $O(r)$ candidate term by $O(nr)$. A cold approach that re-inverts s at every criterion or re-decomposes for each series incurs additional repeated work. Likewise, a banded Cholesky implementation that caches the same K factors across R series may incur $O(Knd^2 + RKnd)$ fitting arithmetic, *before* trace/leverage computation. This is why different setup/reuse policies can reverse a timing winner.

The reference-mean policy changes coefficients of the right-hand side, not the spectrum. Fixed $\mu = 0$ and fixed $\mu = 0.15$ share precisely the same $G_{n,d}$. Estimating $\widehat{\mu}$ from $D_d y$ adds an $O(nd)$ pass per series; the archived comprehensive Windows run did *not*

separately time this data-dependent mean-estimation step.

2.4. What is actually maximized under the Gaussian ML criterion?

The benchmark's label *ML* means profiled Gaussian *marginal* variance-component likelihood, not pointwise maximum likelihood of the fitted curve. It assumes independent observational errors and a Gaussian penalty prior,

$$\begin{aligned} y &= \tau + \varepsilon, \quad \varepsilon \sim N(0, \sigma_\varepsilon^2 I), \\ D_d\tau - \mu\mathbf{1} &\sim N(0, \sigma_\eta^2 I), \quad \lambda = \sigma_\varepsilon^2/\sigma_\eta^2. \end{aligned} \tag{5}$$

The polynomial nullspace is unpenalized and eliminated from the reduced likelihood. On the $r = n - d$ penalized singular coordinates z_j, the marginal variances are $\sigma_\varepsilon^2 c_j(\lambda)$ with $c_j = 1 + 1/(\lambda\delta_j)$. Profiling the unknown observation variance gives

$$\begin{aligned} \widehat{\sigma}_\varepsilon^2(\lambda) &= \frac{1}{r}\sum_{j=1}^{r} z_j^2/c_j(\lambda), \\ -2\ell_{\mathrm{prof}}(\lambda) &= r\log\widehat{\sigma}_\varepsilon^2(\lambda) + \sum_{j=1}^{r}\log c_j(\lambda) + \text{constant}. \end{aligned} \tag{6}$$

This is a REML-like reduced-coordinate likelihood and it *does* depend on a variance model. After selection, one may also compute $\widehat{\sigma}_\eta^2 = \widehat{\sigma}_\varepsilon^2/\widehat{\lambda}$. The benchmark uses fixed $\mu = 0$ or 0.15; any data-derived $\mu = \mathrm{mean}(D_d y)$ would instead constitute a conditional plug-in, *not* joint maximum likelihood over $(\mu, \lambda, \sigma_\varepsilon^2, \sigma_\eta^2)$. For fixed variances and λ, maximizing the Gaussian posterior in τ recovers the ordinary PLS trend; that is a *different* optimization from selecting λ by marginal ML.

2.5. Matched versus unmatched coordinate searches

We separate *native s grid* (invert each s_j during a search), *premapped s grid* (invert once, pass the identical λ_j to the score), and an *independently spaced log-λ grid*. Only the first two evaluate the same statistical candidates, and hence they must agree in exact arithmetic. The third may run faster but selects among *different interior penalties*; its search time alone cannot establish an efficiency advantage at equal statistical quality. Continuous bounded optimizers in s and $\log\lambda$ follow different paths and are local searches, not certified global optima. Exact $s = 0$ and $s = 1$ endpoints are scored analytically.

The classical residual-based BIC expression has a divergent interpolation boundary; our numerical companion studies this limitation. Here BIC timings are purely computational and do not imply the unrestricted criterion is statistically well-posed.

3. Structured end-to-end criterion computation

A fast linear-system solution is not automatically a fast smoothing-parameter selector. For each candidate penalty, an implementation must obtain the trend, its residual sum of squares and, depending on the objective, the effective degrees of freedom (EDF), the diagonal leverage values or the marginal likelihood determinant. These auxiliary calculations are included in our definition of a *complete computational pipeline*. We give a structured construction that avoids a dense $n \times n$ inverse.

3.1. Banded factorization and the selected inverse

Let $A_\lambda = I + \lambda Q_d$, $Q_d = D_d^\top D_d$, and $\lambda \in (0, \infty)$. The symmetric positive definite matrix A_λ has half-bandwidth d. Factor it as $A_\lambda = L_\lambda L_\lambda^\top$ with lower triangular L_λ, whose nonzero entries satisfy $0 \le k - i \le d$. Define $Z_\lambda = A_\lambda^{-1}$. Although Z_λ is generally dense, only the entries Z_{ij} with $|i - j| \le d$ are needed in the selected-inverse recurrence. Backward substitution from $i = n$ to 1 gives

$$Z_{ij} = -\frac{1}{L_{ii}}\sum_{k=i+1}^{\min(n,i+d)} L_{ki} Z_{kj}, \quad i < j \le \min(n, i+d), \tag{7}$$

and

$$Z_{ii} = \frac{1}{L_{ii}^2} - \frac{1}{L_{ii}}\sum_{k=i+1}^{\min(n,i+d)} L_{ki} Z_{ki}. \tag{8}$$

Symmetry supplies Z_{kj} when $k > j$. These are selected-inverse (Takahashi-type) identities (Erisman and Tinney, 1975), not a proposal to store the full inverse. A banded factorization and selected inversion each require $O(nd^2)$ arithmetic, with $O(nd)$ factor and selected-entry storage for a fixed small difference order. The leverage diagonal is $h_i = Z_{ii}$ and $\nu(\lambda) = \sum_i h_i$.

The finite-penalty right-hand side is constructed directly:

$$b_{\lambda,\mu} = y + \lambda\mu D_d^\top \mathbf{1}, \qquad A_\lambda\widehat{\tau} = b_{\lambda,\mu}. \tag{9}$$

This bypasses the potentially huge polynomial particular solution p_μ, avoiding one source of catastrophic cancellation at large d and nonzero μ. It does not, however, guarantee forward accuracy when A_λ becomes badly conditioned. We require finite outputs, residual checks and independent reference comparisons before accepting a timed result.

3.2. Conditional criteria and computational requirements

For a *fixed numeric* μ define $e = y - \widehat{\tau}$, $\mathrm{RSS} = e^\top e$ and $h_i = (H_\lambda)_{ii}$, with $H_\lambda = A_\lambda^{-1}$. The classical leave-one-observation-out criterion and generalized cross-validation are

$$\mathrm{CV}(\lambda) = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{e_i}{1-h_i}\right)^2, \qquad \mathrm{GCV}(\lambda) = \frac{\mathrm{RSS}/n}{(1-\nu(\lambda)/n)^2}. \tag{10}$$

The original residual-based criteria used in the accompanying benchmark have the form

$$\begin{aligned} \mathrm{BIC}_{\mathrm{RSS}}(\lambda) &= \log(\mathrm{RSS}/n) + \nu(\lambda)\log n/n, \\ \mathrm{AICc}(\lambda) &= \log(\mathrm{RSS}/n) + \frac{2\nu(\lambda)+1}{n-\nu(\lambda)-2}. \end{aligned} \tag{11}$$

We retain the exact score conventions of the existing software rather than silently substituting a different information criterion. The BIC expression decreases without bound as $\lambda \downarrow 0$ under a nondegenerate interpolation limit; the finite grid lower boundary must therefore be disclosed. The profiled Gaussian marginal-likelihood criterion additionally requires a weighted quadratic form and a log determinant on the penalized subspace, described in Section 2. Its implementation is not interchangeable with merely calculating a residual MSE.

If μ is estimated from the same observations as $\widehat{\mu} = \mathrm{mean}(D_d y)$, then H_λ is no longer the complete response-to-fit derivative of the plug-in procedure. The standard fixed-μ leverage formula in (10) is *not* an exact leave-one-out refit of this estimated mean. Our planned fixed-mean comparisons distinguish $\mu = 0$ and $\mu = 0.15$; plug-in μ is a separately labeled conditional marginal-likelihood analysis.

3.3. Sparse and iterative methods with honest diagnostic costs

We benchmark nine end-to-end *pipelines*, not nine interchangeable Cholesky routines. Sparse SuperLU, general banded LU, SuiteSparse CHOLMOD, and preconditioned conjugate-gradient (PCG) each provide a fitted trend through a different algorithm. In the current research code their CV/GCV/AICc/BIC/ML scores are completed by an *additional* banded selected-inverse diagnostic step. Thus the hybrid runtime includes both the named fit backend and the added diagnostic factorization. SciPy sparse LU is not CHOLMOD; an unavailable optional CHOLMOD installation is recorded as such.

PCG uses a matrix-free application of $I + \lambda D_d^\top D_d$ and FFT-periodic, DST-Dirichlet or Jacobi preconditioning. The FFT and DST symbols approximate the natural finite-series boundary structure; they are not exact diagonalizations, and no fixed small iteration count is guaranteed. Nonconverged PCG runs cannot enter the successful-time comparison.

4. Experiment006: factorial evaluation protocol

The present expansion treats *complete selection time* as its principal response. Statistical estimation error, including trend RMSE, is outside this experiment. Numerical validity remains an eligibility condition: a faster output is not a successful computation if it fails the stated mathematical equations. We distinguish the *prospective* Experiment006 design from archived small-n observations reported in Section 6. This section describes a partially implemented research protocol; the large factorial has not been measured.

4.1. Four crossed workloads

Write M for the number of candidate smoothness values, R for independent series sharing (n, d), and C for the number of criteria evaluated. Four workload classes cross the two relevant dimensions: The first workload exposes cold setup costs; the second exposes amortization across penalties; the third isolates reuse across distinct observed vectors; and the fourth measures a realistic multi-series selection batch. In particular, a grid-level speedup is not extrapolated from timing one isolated factorization.

4.2. Factors, endpoints and numerical scales

Because $S_d(0) = 0$ and $S_d(+\infty) = 1$, the endpoint $s = 1$ is a *constrained* PLS limit, not a very large finite penalty. Any unstable high-order polynomial projection at this endpoint is recorded as a numerical failure. At $n = 131072$ a single dense float64 $n \times n$ array requires 128 GiB before factors or workspaces; dense inversion and full SVD must be explicitly marked resource-ineligible at that scale, not dropped from the design.

Small and large observed amplitudes alone do not define the numerical problem: fixing a nonzero μ while rescaling y changes the relative strength of the imposed difference-mean constraint. The factorial therefore crosses amplitude and μ rather than treating their scales as exchangeable. With $d = 6$ the nonzero-μ limiting polynomial can have very large coefficients even when the data values are small.

4.3. Fair coordinate comparison

Define a common candidate set $\mathcal{G} = \{s_1, \ldots, s_M\}$ and its image $\Lambda_{n,d}(\mathcal{G})$. In the on-demand implementation, s_j is inverted whenever requested. In the pretranslated implementation all λ_j are computed once and passed directly to the same backend criterion. Therefore,

$$F_{\mathrm{direct}}(s_j) = F_{\mathrm{translated}}\big(\Lambda_{n,d}(s_j)\big) \tag{12}$$

in exact arithmetic for any one criterion F and fixed reference mean. Identical selected values (within numerical tolerance) are a *correctness check*; relative time

Table 2
Leading arithmetic complexity for one finite-penalty, fixed-μ complete criterion evaluation with difference half-bandwidth d and $r = n - d$. For hybrid implementations, additional banded diagnostics are *charged*, even if their fit operation is inexpensive. q is the attained PCG iteration count; these are arithmetic orders, not measured timings.

Complete pipeline	Preparation / factor	Criterion	Memory
Fresh dense inverse	$O(n^3)$	$O(n^2)$ after inverse	$O(n^2)$
Full SVD (after one decomposition)	$O(nr^2)$ once	$O(r)$ for GCV/AICc/BIC/ML; $O(nr)$ for CV	$O(nr)$
Banded Cholesky + selected inverse	$O(nd^2)$	$O(nd)$ for solve and criterion	$O(nd)$
Banded or sparse LU + banded trace	fit-backend dependent $+O(nd^2)$	$O(nd)$ after diagnostics	fit memory $+O(nd)$
CHOLMOD + banded trace	sparse-factor dependent $+O(nd^2)$	$O(nd)$ after diagnostics	sparse-factor memory $+O(nd)$
PCG + banded trace	$O(qnd)$, or $O(qn(d + \log n))$ with transforms, $+O(nd^2)$	$O(nd)$ after diagnostics	$O(nd + n)$, excluding FFT workspaces

Table 3
Four full-pipeline workloads. A single S is 0.5; the complete grid uses the levels in Table 4. A criterion score, not merely a fit, is timed.

Workload	M	R
Single value, single series	1	1
Grid, single series	$M > 1$	1
Single value, multiple series	1	$R > 1$
Grid, multiple series	$M > 1$	$R > 1$

quantifies reuse of conversion, not a change of statistical estimand. Independently spaced log-λ grids sample different models and require a separate sensitivity analysis.

Let T_0 be solver preparation, T_Λ the once-computed grid mapping, T_p the data projection for one series, and T_F one criterion score. For a spectral GCV-like objective a stylized cost model is

$$T_{\text{shared}} \approx T_0 + T_\Lambda + RT_p + RCMT_F, \tag{13}$$

whereas repeated on-demand inversion can introduce up to RCM mapping calls. These are accounting models, not measured speedups. CV changes T_F through leverages; sparse and hybrid algorithms incur separate factorization, cache and diagnostics costs. A fair comparison fixes (n, d, μ, R, M), criteria and candidates, then records preparation, mapping, per-series projections, candidate scoring, total time and retained memory.

4.4. Paired measurement and analysis

Each run records source revision, CPU, operating system, Python/SciPy/BLAS versions, thread counts, warm-up policy, candidate count, random seed and every attempted computation. Independent seeded timing blocks with counterbalanced method order are the proposed publication protocol. Only complete, numerically valid attempts enter winner comparisons. Planned summaries include means, medians, standard deviations, confidence intervals by independent timing block and paired speedups against explicit dense inversion wherever feasible. For fixed levels of other factors, a regression $\log T = \alpha + \beta \log n$ across at least three series lengths gives a descriptive empirical scaling exponent, not a proof of big-O complexity.

The runnable design manifest, bounded slices and mathematical tests reside in the separate Experiment006 research branch. The largest $M = 100000$ by $n = 131072$ combinations require resource gating, not extrapolation from completed pilots. Incomplete and ineligible cells are scientifically meaningful observations, not measured timing winners.

5. Archived benchmark design and provenance

This section analyzes *completed and committed historical* timing experiments; the prospective Experiment006 factorial in Section 4 has different axes and is not claimed as measured. It does not substitute fitted complexity curves for measured time or combine machine-specific durations from different hosts into a single speedup. The principal dataset was collected on the author's Windows 11 workstation (the environment identifies Windows build 26100) with Python 3.11.17, NumPy 2.4.6, SciPy 1.17.1, OpenBLAS limited to one thread, two experimental seeds and seven timing repetitions per reported configuration.

The complete Windows experiment contains 80,640 raw timing records and a separate computing-environment record; precise repository paths and checksums are

Table 4
Target axes of Experiment006. This is a *planned design*, not completed measurements. Each new smoothness grid includes $s = 1$ but starts above zero; historical 17-point grids also included $s = 0$.

Axis	Levels / interpretation
Series length n	$64, 128, 256, 512, 1024, 16384, 131072$
Difference order d	$1, 2, 3, 4, 5, 6$
Reference mean	Fixed $\mu = 0$ or $\mu = 0.15$; observed-difference plug-in $\widehat{\mu}$ separately, conditional ML only
Observation magnitude	Low (10^{-3} times a fixed seeded shape) and high (10^{6} times the same shape)
Criterion	Conditional CV, GCV, AICc, residual-based BIC, profiled Gaussian marginal ML
Smoothness grid	$\{j/M : \ j = 1, \ldots, M\}$, $M = 100, 1000, 10000, 100000$, including $s = 1$ exactly
Selection coordinate	On-demand $s \mapsto \lambda$ inversion; once-pretranslated $\lambda_j = \Lambda_{n,d}(s_j)$ and identical candidate scoring
Complete pipeline	Naive inverse, SVD, banded Cholesky/selected inverse, and six named LU/CHOLMOD/PCG plus banded-diagnostic hybrids
Reuse	$R = 1$ or $R > 1$ independent series; independent paired timing blocks, cold/warm accounting and cache-memory constraints

Table 5
Actual coverage of the archived timing studies. The principal full-factorial rows are *not* 80,640 independent trials: they comprise 40,320 design cells and two recorded seeds. Additional repetitions are timed measurement blocks. This table separates which extra data would still be required to claim all requested combinations.

Corpus	Measured design	Limitation
Full Windows search	$n = 24, 48$; $d = 0, \ldots, 5$; $\mu = 0, 0.15$ fixed; $R = 1, 3, 8$; CV/GCV/AICc/BIC/ML; seven backends; eight search policies; cold/shared setup; 2 seeds; 7 repetitions	No $d = 6$ timing or empirical $\widehat{\mu} = \mathrm{mean}(D_d y)$
20-seed solver study	SVD, dense eigendecomposition, banded Cholesky; $M = 1, 16, 64$ full curves; $n = 32, 96, 192$ plus selected $d = 1, 3, 5$ cells; 20 independent seeds	Fit-only, not five-criterion selection
GCV reuse study	Six solver families; uniform-s versus log-λ grid; fixed means 0 and 0.15; varied grid sizes and series reuse	GCV only; generally different grid interiors
Extended fit/trace studies	Dense fresh/cached, spectral, general banded and Cholesky alternatives; selected n up to 400, M up to 256	Earlier machine/candidate regimes; not pooled with Windows factorial

listed in the accompanying `SOURCE_AUDIT.md`. A checked analysis yields 40,320 two-seed design cells and verifies exactly 4,484 matched native-s/premapped-grid valid seed pairs with *zero difference* in selected smoothness and score. The associated `windows_full_s_analysis/` folder preserves checks and per-cell paired costs. We use only cells whose complete evaluated grid passed the backend's recorded numerical validation; an implementation that returned a partial grid does not win a speed ranking.

5.1. Solver families and workloads

The seven Windows backends are: fresh dense solve, cached-Q dense solve, general banded solve, dense Cholesky, banded Cholesky, eigendecomposition of Q_d and full SVD of D_d. The five criteria are classical fixed-μ CV, GCV, AICc, residual-based BIC and the implemented conditional marginal-likelihood score. Eight policies cover one point at $s = 0.5$ with native/premapped/direct penalties; complete 17-point native-s, premapped and independent log-λ grids; and bounded *local* optimization in s and $\log \lambda$. All policies are evaluated at $R \in \{1, 3, 8\}$ and cold/shared preparation. Grid conversion is separately timed; a standalone premapped search charges it once, while reuse across criteria charges it only once across the entire joint workflow.

The full normalized-s grid includes the exact $s = 0$ and $s = 1$ endpoints plus near-one finite points through $s = 1 - 10^{-12}$. For large d these correspond to potentially extreme finite penalties; failed factorizations and numerically inaccurate criterion values are explicitly tracked. The Gaussian likelihood used in this corpus was marked *not applicable* for $d = 0$. Although that historical run includes order zero, all smoothness theory and the proposed new methodological scope in this manuscript use $1 \leq d < n$.

For the representative timing table, the original script selected $n = 48, d = 2, \mu = 0, R = 1$, SVD, and the median over its two Windows seeds. The log-λ grid has 17 points but different interior values, and the continuous policies are not guaranteed global optimizers. No inferential result follows from comparing differently discretized search spaces.

Table 6
Five completed 17-point selection searches for $d = 3$, $n = 48$, fixed $\mu = 0$, SVD, $R = 1$. Values are two-seed medians in milliseconds. Both policies evaluate the identical full $S \in [0, 1]$ grid; premapped standalone time includes conversion once per criterion.

Criterion	Repeated S inversion	Premapped S grid
CV	2.058	1.846
GCV	1.958	1.723
AICc	2.024	1.724
Residual BIC	1.972	1.724
Profiled Gaussian ML	2.038	1.776

5.2. Statistical interpretation of runtime measurements

A two-seed Windows timing median describes that host and work unit; it is *not* a sample of 80,640 independent processors or experiments. The separate solver study used 20 independent data seeds and paired timings to compute pointwise confidence intervals and Holm-corrected sign-flip tests for prespecified fitted-trend workloads. We report these uncertainty summaries separately rather than apply the study's p-values to the different Windows factorial. Setup, grid preparation and criterion scoring have different accounting boundaries, so their elapsed times must not be summed or divided without matching definitions.

The archived sources also do not contain an end-to-end, all-criteria $d = 6$ timing factorial, nor an additional $\widehat{\mu} = \mathrm{mean}(D_d y)$ timing policy. These are concrete extensions for the final combined paper, not silently filled gaps. The measured numerical-error companion covers $d = 6$ but cannot be converted into a runtime observation after the fact.

6. Measured computational results

6.1. One full S-grid, with or without reusable inverse mapping

The principal case now uses $n = 48, d = 3, \mu = 0, R = 1$, the SVD backend and five selection criteria. Both policies evaluate *exactly the same 17 normalized-smoothness candidates*, including $S = 0$ and $S = 1$: native-S repeatedly invokes $\Lambda_{n,d}$, whereas the cached policy translates the grid once and reuses its penalties. This is an equal-estimator computational comparison, not a contest between different statistical grids. The recorded two-seed median timing values are inserted automatically from the original Windows CSV, not manually estimated. Although the candidate grid includes $S = 0$ and $S = 1$, some criterion values at those endpoints can be mathematically undefined (notably residual BIC at interpolation), and those scores are not reinterpreted as admissible optima.

In the jointly reused five-criterion $d = 3$ workflow, the native-S total was 10.050 ms, versus 3.767 ms for one shared mapping plus five backend-only searches (ratio 2.668). In contrast, the premapped column of Table 6 charges the mapping once *per standalone criterion*; summing those values would overcount shared preparation.

For a second, historically reported $d = 2$ comparison at the same $n = 48, \mu = 0, R = 1$, the archived five native-S searches total 11.234 ms, versus 3.980 ms when the grid is mapped once and shared across five backend-only searches: a descriptive factor of 2.823. These $d = 2$ values are *not* relabeled as $d = 3$ timings; the new $d = 3$ table and plot use their own archived records. Within each difference order, a precomputed grid can be shared across criteria, series and fixed μ values without altering the PLS estimator.

6.2. One evaluation, full grid and repeated independent series

Figure 4 plots all seven backends at a *single* $s = 0.5$ GCV score, one mapped 17-point GCV grid and eight independent full-grid GCV searches, under fixed $n = 48, d = 3, \mu = 0$. It uses the original full-factorial Windows rows, drawing a timing only where both experimental seeds completed that particular workload with valid numerical scores. Unsuccessful or incomplete cases are missing, not assigned fictitious times. The figure makes visible why a one-evaluation ranking need not generalize to selection or reuse workloads: decomposition cost is paid once, while data projections, curve solves and per-candidate trace work scale differently across algorithms.

6.3. Repeated grid reuse and difference order

Figure 5 shows the recorded ratio of cold to shared backend-only time for $R = 8$ complete s-grid searches across $d = 1, \ldots, 5$, conditional $\mu = 0$, $n = 48$, and the SVD backend. Differences among criteria are attributable to distinct scoring operations and reuse boundaries, not a change in statistical PLS formulation. A ratio greater than one favors shared setup for that cell; outcomes missing the necessary two valid seeds are left blank, not silently imputed.

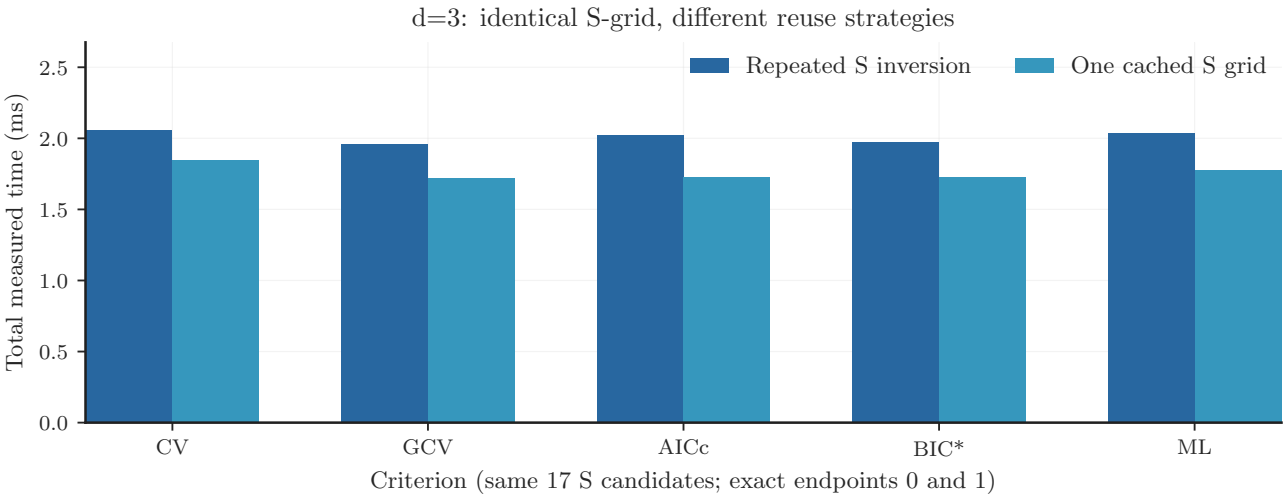


Figure 3: Full $S \in [0,1]$ selection at $d = 3$, $n = 48$, fixed $\mu = 0$ and SVD, with five criteria: repeated on-demand $S \mapsto \lambda$ inversion versus one cached candidate translation. Both policies evaluate identical penalties. Bars are observed two-seed median standalone timing results, including one-time mapping cost in the second policy.

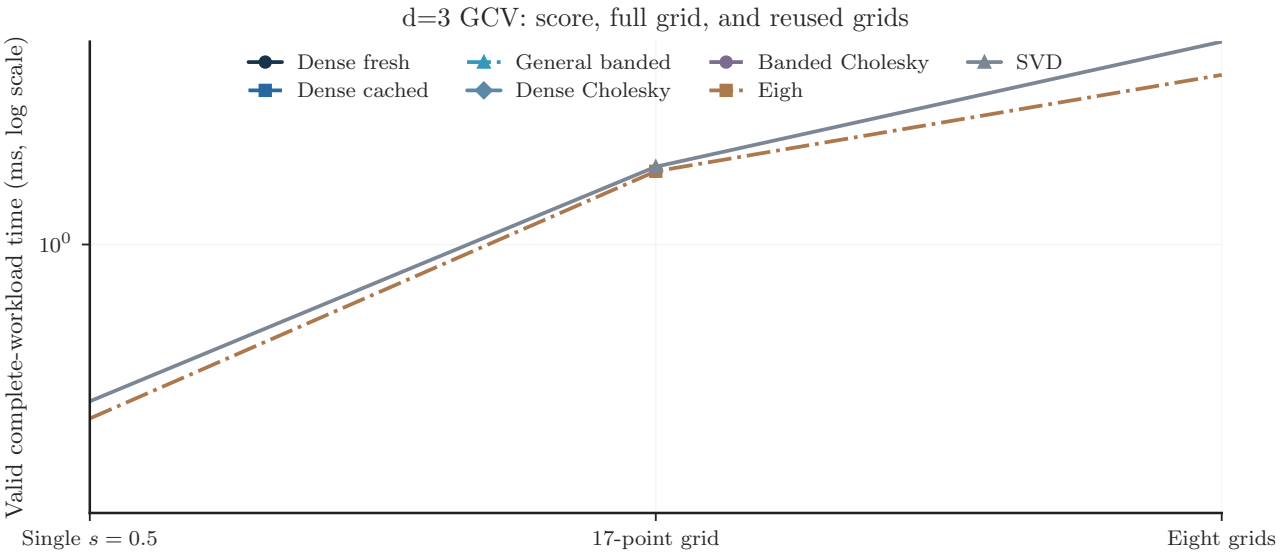


Figure 4: Observed standalone elapsed time for one GCV score ($s = 0.5$), one 17-point grid, and eight independent grids, across seven implementations. Data are from the two-seed Windows full factorial with $n = 48, d = 3, \mu = 0$ and shared preparation. All seven methods supplied a valid single-score evaluation, but only two supplied valid complete full-grid and repeated-grid cases at this difficult $d = 3$ setting. Points are shown only for workloads with complete numerically valid seed pairs; missing points are not zero elapsed time or a performance win. The three workload classes are distinct algorithms, not repeated scalar multiplications of one measured time.

6.4. Longer series and repeated fitted curves

The separate complete-trend benchmark, with n extending to 400 and up to 256 repeated fits, documents substantial changes in winners. For $n = 400, d = 2, \mu = 0$, the ratio of fresh dense to spectral total time was 0.158 at $M = 1$ but 28.314 at $M = 256$; 0.158 means the dense approach was faster for the single fit, whereas 28.314 means the spectral implementation was faster than the repeatedly fresh dense solver on the larger sweep. Against a properly general banded solver the corresponding ratios were 0.055 and 0.339, so *banded remained faster* in those two recorded workloads. These are host-specific paired cases, not universal crossovers and not five-criterion selections.

The independent 20-seed solver study also supports comparing higher orders. Figure 6 shows the original paired geometric-mean SVD/banded-Cholesky total-runtime ratio at $n = 96$, $M = 16$ complete fitted curves, fixed $\mu = 0$, and orders $d \in \{1, 2, 3, 5\}$. There is *no*

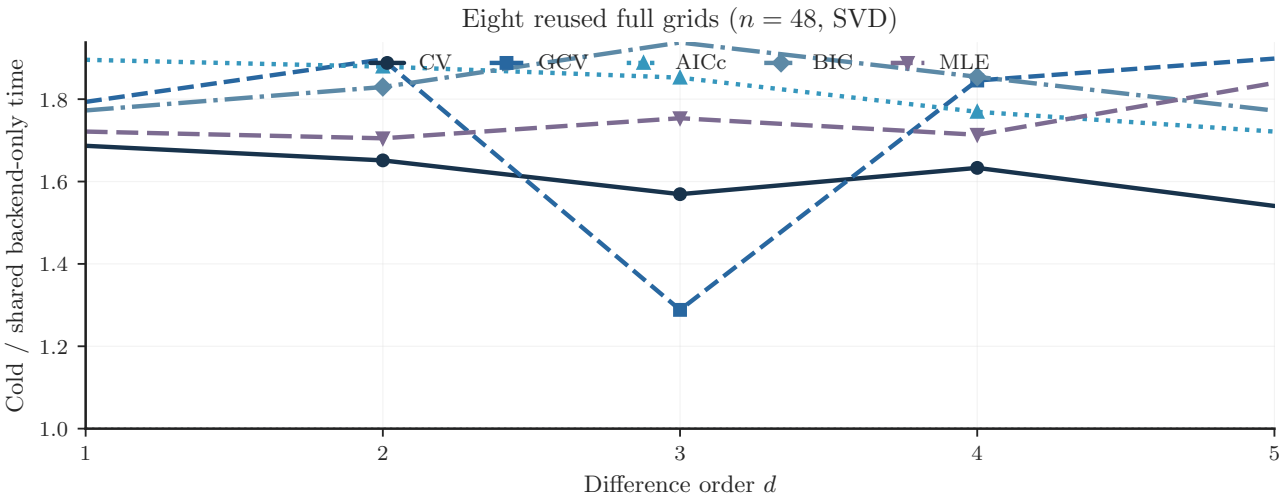


Figure 5: Cold versus shared implementation cost for eight repeated matched-grid selections by difference order and criterion. A vertical value above one indicates less backend-only time with shared preparation. $n = 48$, fixed $\mu = 0$, SVD; exact source is the archived two-seed Windows timing CSV. No $d = 6$ Windows timing was recorded.

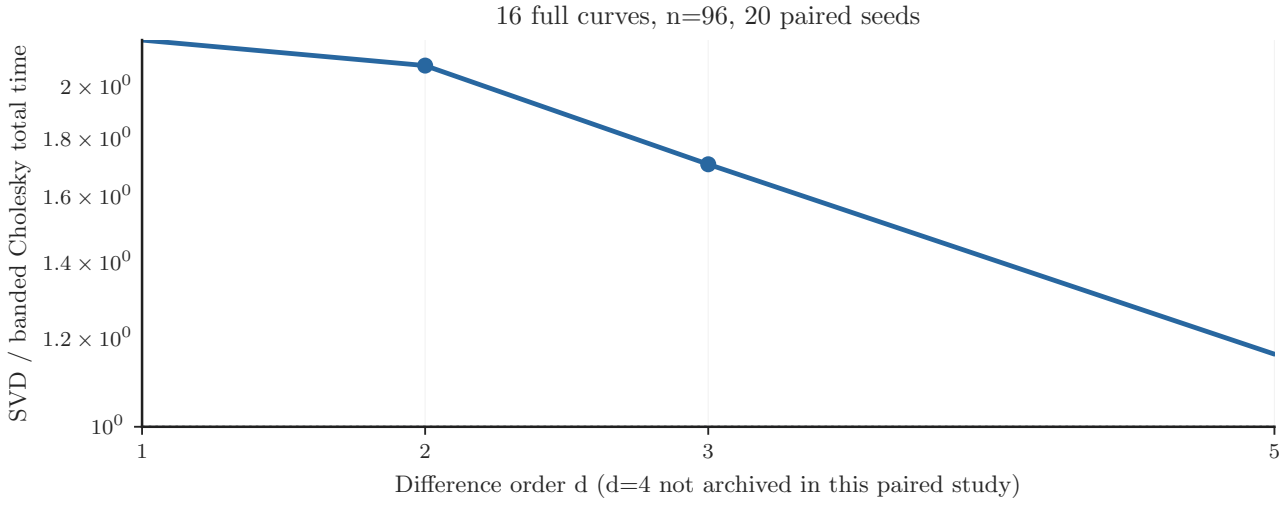


Figure 6: Measured paired SVD/banded-Cholesky full-curve total-time ratios across available difference orders at $n = 96$, $M = 16$ and $\mu = 0$. Each order has 20 independent paired data seeds. $d = 4$ was not included in this matched solver workload; no result is extrapolated. This is not a criterion-selection benchmark.

paired $d = 4$ observation for this exact workload in that source: the figure does not interpolate or fabricate one. A ratio above one favors banded Cholesky and below one SVD. These are *fit-only* timings and do not cover trace, leverage or five-criterion selection.

6.5. Exact matched-grid equivalence

The complete source audit verified 4,484 numerically valid seed-matched native-s/premapped-λ grid comparisons with *zero* differences in selected s and scored values. This is expected from (2): a monotone coordinate transform does not change the statistical model. The computational advantage lies in *the caching policy*, not statistical changes. Results from unmatched log-λ or bounded continuous search cannot establish this equivalence because they evaluate different candidate sets.

No measured cell in this section should be considered an objective benchmark at $d = 6$ or for data-derived $\widehat{\mu}$; those remain future research dimensions.

7. Evidence status and pre-registered comparisons

This section fixes the separation between historical measurements and the prospective experiment. The archived Windows timing corpus in Section 6 covers $n = 24, 48$ and $d \leq 5$, whereas Experiment006 expands

Table 7
Evidence classification as of this manuscript version. A green software test or successful small pilot does not establish the full-factorial timings. In particular, the only historical joint five-criterion timings here are from $n = 48$.

Question	Available evidence	Remaining evidence needed
Matched S versus translated-λ grids	4,484 valid matched two-seed cases in historical Windows source; equal selected candidates and scores	All requested M, n, d, fixed-μ and amplitude regimes
Five-criterion reuse across the same mapped grid	Historical $n = 48, d = 2, 3$, fixed $\mu = 0$, SVD cases	Repeated criteria and series at large n including nonzero μ
Single fit versus repeated fits	Historical full-fit sweeps and separate 20-seed solver study	Direct full-criterion comparisons with the same candidate grids and hardware
Sparse, banded and iterative complete criteria	Implemented reference pipelines and mathematical parity tests in Experiment006 research code	Repeated, paired end-to-end timings and numerical validation on PCAG103A-08
Maximum smoothness $S = 1$ and $d = 6$	Analytic limit and selected-inverse verification for small n	High-order independent accuracy checks at larger n and high data magnitude
Asymptotic scaling vs measured exponents	Derived arithmetic/storage bounds	Three or more measured n levels per comparable factor cell, confidence intervals and clear resource-ineligible statuses

to seven series lengths and all six positive difference orders. The two data sources must not be pooled into one purported factorial cell or used to construct an unmeasured empirical complexity exponent.

Our first hypothesis is that a banded Cholesky plus selected-inverse pipeline will avoid the quadratic memory and cubic setup costs of full dense spectral or inverse-based approaches when d is fixed. The second is that full SVD will become competitive in configurations with enough repeated candidate scores to amortize its initial decomposition. The third is that one-time translation of a fixed S grid will reduce mapping overhead across independently observed series, with identical selected S values for complete, valid matched grids. The fourth is that higher orders, strong smoothness and nonzero reference means can change the numerical failure pattern enough that a raw elapsed-time ranking becomes misleading without an accuracy gate. These are *testable research hypotheses*, not conclusions extrapolated to unmeasured factor levels.

For the full factorial, the primary observed response is end-to-end time in seconds, conditional on numerical correctness. Secondary responses are setup time, one-time mapping time, score time, peak memory, number of root solves, number of PCG iterations and the fraction of ineligible or failed attempts. When the naive dense baseline cannot run, a speedup against that baseline is *undefined*; it must never be replaced with a synthetic time predicted from $O(n^3)$. Mean and confidence intervals are reported across independent paired blocks rather than treating tightly correlated inner-loop stopwatch samples as independent observations.

This distinction also guides manuscript revision. Future Experiment006 results will be inserted only after their raw timing CSV, environment metadata, numerical validity summary, and source revision have been committed to the repository. Until then, the present document is a mathematically developed *working draft* with a completed historical benchmark, not a completed seven-length computational study.

8. Discussion: when does a reusable smoothness interface pay off?

The main distinction is between *a mathematical coordinate* and *a computational policy*. The normalized smoothness map S_d is an exact reparameterization for any $1 \le d < n$ with fixed numeric μ. It is valuable to users because the range and direction of smoothness remain interpretable across d even though penalties differ radically in magnitude. Nothing in this argument implies that optimizing in s must take less time than in λ; in fact, evaluating $s \mapsto \lambda$ at every optimizer call introduces extra computation. The measured advantage appears when the same candidate grid and spectral information are reused across calls.

Figure 3 and Table 6 use $d = 3$ and demonstrate that native and premapped full-S searches have identical statistical candidates. An independently sampled log-λ grid was included in the original experiment solely as a conventional computational control; because it selects among different interior smoothness levels, we remove it from the main display rather than treating it as a competing interpretable interface. Bounded optimizers likewise take different search paths and do not certify global optimality.

The best numerical backend depends on the *actual requested output*. A banded factorization can be very economical for one fitted curve, whereas amortizing one dense SVD can accelerate repeated spectral GCV/ML scores. Exact CV leverages and specialized trace computation can reverse rankings yet again (Weinert, 2007; Biessy, 2026). High-s, large-d conditioning matters too.

The independent accuracy companion found that full SVD with shift-free reconstruction is especially reliable in the tested extreme regime. However, that finding should not be back-projected as a universal wall-time speed ranking, nor should older valid low-s benchmark cells be discarded.

8.1. What the historical factorial does not establish

Although the archived Windows source is a complete factorial *within its declared axes*, it covers $d = 0, \ldots, 5$ rather than $1, \ldots, 6$ and only fixed $\mu = 0$ versus fixed $\mu = 0.15$. It does not test $\widehat{\mu} = \mathrm{mean}(D_d y)$ as a separately timed policy. Nor does a two-seed, $n = 24, 48$ matrix establish asymptotic constants or algorithmic scaling laws at arbitrary n. We therefore report the big-O analysis independently of the machine measurements. Future work should add $d = 6$, data-derived and jointly estimated reference means, wider n, additional independent timing sessions and statistical evaluation of complete-workflow validity.

For a truly complete cross-backend selection comparison, the selected-inverse/trace approach of Weinert (2007) is essential prior art. Additional workloads should distinguish an observed-difference plug-in from fixed μ, conditional LOOCV from chronological forecast validation, independent-λ from matched-s candidate grids, and numerical failures from statistically undefined criterion values. The residual-based BIC interpolation limit should be restricted explicitly if included as a selector, as detailed by the companion reliability study.

8.2. The four-workload computational frontier

The prospective design explicitly crosses one versus many S values with one versus many independent series, under every difference order $d = 1, \ldots, 6$ and fixed zero or nonzero mean of differences. This corrects an important limitation of fit-only solver comparisons: the optimal computational policy can depend on whether factorization, smoothness inversion, traces and leverage diagonals can be reused. A method must remain numerically admissible to appear in the fastest-time ranking, and a hybrid pipeline must pay the complete cost of its extra diagnostics. Near $S = 1$, exact polynomial limits and high-order error checks are essential, especially with nonzero μ.

At the largest n and grid levels, some matrix algorithms or caches are not feasible. That is an algorithmic finding to be documented by theoretical bounds and explicit resource status, not a missing bar in a runtime figure. The most informative output will therefore be a conditional computational frontier by $(n, d, \mu, M, R, \text{criterion})$, together with memory consumption and failure rates, rather than a declaration of a universally optimal solver. Any empirical scaling exponent will be estimated only from genuinely measured, method-matched cells.

8.3. Toward one combined manuscript

We propose ultimately combining: (i) the mathematical $s \leftrightarrow \lambda$ bijection and fixed/nonzero-μ formulas; (ii) the present algorithmic cost models, matched-coordinate parity and amortized runtime evidence; and (iii) independent high-precision solver validation and finite-sample latent-trend RMSE from the numerical companion. The comparison should use a *joint accuracy–time admissible frontier*: a solver must deliver valid equivalent criterion values over the *complete* candidate set before its timing is ranked. This avoids favoring fast, inaccurate or incomplete high-order factorizations and clarifies whether a cached spectral default is the best robust option for a specified workload.

9. Conclusions

A normalized smoothness index is an established, mathematically interpretable interface for quadratic finite-difference PLS at any $1 \leq d < n$, independent of the fixed mean of differences. The full singular spectrum makes the map invertible, permits reusable candidate grids and enables amortized criterion scoring. The revised Windows analysis compares an identical complete S grid at $d = 3$, measuring 10.050 ms for repeated native inversion versus 3.767 ms with one cached grid across five criteria (2.668-fold). The historical $d = 2$ case separately gives a 2.823-fold reduction. Neither ratio is a universal speedup, and only two backends completed the difficult $d = 3$ full-grid GCV tasks validly. Other complete-fit benchmarks show why banded solvers remain valuable, and no universal fastest backend follows. The present expanded manuscript supplies a structured-criterion complexity analysis and a fully specified four-workload protocol, while the large Experiment006 results remain pending. The historical timing observations cannot by themselves establish the fastest full-criterion pipeline for $d = 6$, nonzero mu and large n.

Data and code availability

Immutable original timings, seed-level results, validations, scripts and environment metadata reside under https://github.com/heri-espino/trendestimation/tree/main/paper/benchmarks/. The original source paths and their restrictions are catalogued in this manuscript's `SOURCE_AUDIT.md`. This paper and its TeX-native figures are maintained together in the repository's computational-costs manuscript directory. No archived numerical results are overwritten or reclassified.

Declarations

Provisional authorship and affiliation are shown for drafting only. Author order, funding, conflicts and contributor statements require author approval before submission.

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